Qien Jing

École polytechnique (Polytechnic Institute of Paris), Rte de Saclay, 91120 Palaiseau, France
Tel: +86 13043330095 or +33 07 43 66 38 49
Email: qien.jing.03@gmail.com or qien.jing@polytechnique.edu

Education

University of Science and Technology of China (USTC) Sept. 2021 - Jul. 2025
B.S. of Physical Science, Senior year
Sub-Major: Plasma Physics
Overall GPA: 3.45/4.3

École polytechnique (Polytechnic Institute of Paris) Sept. 2025 - Present
Master 1 of General Physical

Research Experiences

Research Assistant Jul. 2023 - Jul. 2025
Advisor: Prof. Huishan Cai (Department of Plasma Physics and Fusion Engineering, USTC)

Synergistic Effect between Radial Electric Field and Magnetic Shear on Ion Temperature Gradient Mode
University of Science and Technology of China, Hefei, Anhui, China Nov. 2023 - Nov. 2024
Advisor: Prof. Huishan Cai (Department of Plasma Physics and Fusion Engineering, USTC)

- Discovered that the stabilizing effect of the E_r shear on ITG mode on EAST depends on the sign of the E_r shear, which is different from the results on CBC configuration tokamak. Designed a verification simulation which verified my hypothesis the difference arises due to negative magnetic shear in EAST.
- Found that the ITG mode is more unstable under the positive magnetic shear, but the stabilizing effect of the E_r shear is more pronounced under the positive magnetic shear. Explained the results through the inclination of mode structures.
- Discovered that the global effect of the E_r shear still influences the ITG and and there may be positive feedback between formation of ITB and formation of E_r, which might be the key to the long-time existence of ITB.
- Proposed a mechanism to the experimental E_r manipulation, to suppress the ITG mode effectively

Lost Region Identification in Fast Ion Phase Space in W7-X Stellarator Jul. 2024 - Dec. 2024
University of California, Irvine, CA, USA
Advisor: Prof. Zhihong Lin (Department of Physics and Astronomy, UCI)

- Discovered that the dominant lost region in fast ion phase space is the region of minimum magnetic field amplitude and that the mechanism is ripple trapped particles-induced loss.
- Demonstrated that E_r significantly impacts particle loss: the electron root E_r induces strong poloidal rotation, making deeply trapped particles the dominant loss channel, while the ion root E_r cause the deeply ripple trapped particles that are lost tend to be initially located near the out midplane rather than lowest B field region.

– Future work is exploring the self-consistent E_r that can be generated by injection and consider the impact of self-consistent E_r on the lost particles, which is already in progress.

Publications

- Synergistic effect between radial electric field and magnetic shear on ITG mode, Qien Jing et al 2025 Nucl. Fusion 65 106037.
- Lost region identification in the fast ion phase space on the W7-X stellarator, Q.Jing, E.Green, X.Wei, H.Cai, Z.Lin. (To be submitted)

Research Interests

- Magnetic controlled fusion, Plasma instabilities and turbulence transport

Awards & Honors

- 2023 Excellent Student Scholarship –Bronze (This award recognizes students in the top 20% of their class in the annual comprehensive evaluation)
- 2023 Thesis competition of the Atomic Physics Course (This award recognizes group in the top 30% of the annual competition)
- 2024 Excellent Undergraduate Research Project (This award recognizes students in the top 5% of the annual Undergraduate Research Project)
- 2025 Excellent Undergraduate Thesis (This award recognizes students in the top 5% of the annual Undergraduate Thesis Defense)

Skills

Programming: Fortran, MATLAB, Python, C, Mathematica, Latex

Language: English - TOEFL 91 (Best Score) (R: 29; L: 21; S: 20; W: 21)

Extracurricular Activities

Science and Technology Week: Volunteer guide leader Apr. 2023, May. 2024, May. 2025